POLARON TRANSPORT IN OPTOMECHANICAL ARRAYS

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ABSTRACT

Optomechanics involves quantum-mechanical coupling between electromagnetic radiation and micro (or nano) mechanical motion. A paradigmatic set-up is a cavity consisting of a movable end mirror and a fixed input mirror. A newer development in the field is optomechanical arrays, regular structures of discrete sites, each acting like a cavity, which can carry both vibrations and photons. We consider a one-dimensional chain of optomechanical cavities forming an array. We assume no external laser driving; instead, we take a single photon in the optomechanical cavity with a single identical mechanical frequency per cavity. Photons and phonons are coupled within each cavity and, different sites are coupled through photon and phonon hopping. The platform represents a quantum simulator for polaron dynamics with all these features. A polaron is a quasi-particle formed when a quantum excitation (e.g., charge) moves slowly through a lattice, causing distortion or polarization, and this deformation moves along with the charge. The Hamiltonian of the optomechanical arrays resembles the Holstein Hamiltonian for the polaron. In our context, we consider the eigenstate of the single optomechanical site as polaron and check whether this eigenstate can transport to neighboring sites as a whole or falls apart when both phonon and photon tunneling is enabled. We find that it can stay together, and have numerically simulated polaron transport for a chain of seven cavities. We also obtained the analytical polaron tunneling period and checked its dependence on the photon-phonon coupling and inter cavity tunnelings to better understand the dynamics.

LIST OF SYMBOLS OR ABBREVIATIONS

$\hbar$	reduced Planck's constant
$\hat{H}_{OM}$	single-cavity optomechanical Hamiltonian
$\hat{H}_{hop}$	hopping Hamiltonian
$\hat{H}_{int}$	interaction Hamiltonian
ω	resonant photon frequency
Ω	resonant phonon frequency
g	photon-phonon coupling
J	photon tunneling amplitude
K	phonon tunneling amplitude
$\mid \alpha \rangle$	coherent state
E_p	polaron binding energy
E_{GS}	ground-state energy of single optomechanical cavity
T	polaron tunneling period
J_{eff}	effective photon tunneling
N_{cavity}	number of optomechanical cavities
XMDS	Extensible Multi-Dimensional Simulator
NMQSD	Non-Markovian Quantum State Diffusion

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1. INTRODUCTION

1.1 Motivation

Charge or energy transport affected by an environment coupling is ubiquitous in nature, occurring during photosynthesis in molecular aggregates [1], charge transport in conjugated polymers and conductors [2]. Usually, the environment is modelled as harmonic oscillators, and the charge couples to the vibration modes of the environment called phonons through a system-environment coupling. The exciton-phonon interactions are generally described by the Holstein model [3, 4].
Transport phenomena are relevant in many systems but can not be usually seen in situ. Therefore mapping that to a quantum simulator [5, 6] might give additional experimental platforms for condensed matter theories and give insights into how to improve, guide or control transport. The main idea of quantum simulation is to map it on more accessible platforms with a tunable Hamiltonian. To this end, one can consider quantum simulations of the charge transport on various platforms like Rydberg aggregates [7], cold polar molecules [8, 9], ultracold impurity atoms immersed in a Bose- Einstein condensate [10], and trapped ions [11].

An optomechanical array is a human-made platform that exhibits transport coupled to vibrations in a very different setting [12, 13]. In optomechanical set-up, quantized degree of freedom for the light and the mechanical vibration is coupled via radiation pressure [14]. Many such cavities are, in turn, coupled together in optomechanical arrays such that light and vibration can tunnel from site to site. The Hamiltonian describing these arrays

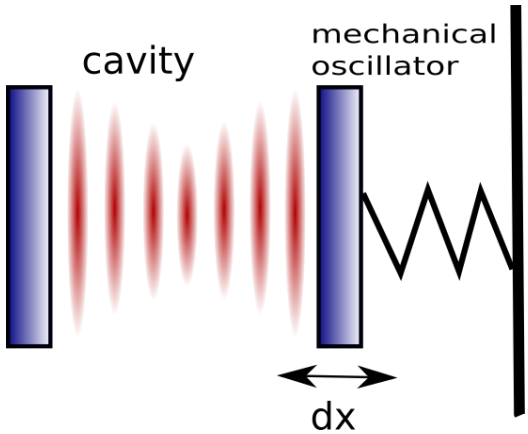

Fig. 1.1: A general single-site optomechanical cavity with a fixed and a movable mirror. The light beam (red) interacts with mechanical oscillator via radiation pressure.

resembles the one describing charge transport. The aim of this project is to explore whether optomechanical arrays are a promising platform for quantum simulations of such transport processes.

1.2 Optomechanics

Light carries momentum that gives rise to radiation pressure, and the interaction between light and matter involves an exchange of momentum between light and a mechanical degree of freedom. This interaction between electromagnetic radiation (photons) and mechanical motion (phonons) via radiation pressure forms the basis of the field of optomechanics. When the light beam is shot inside the optomechanical cavity, it makes several round trips, and when it hits the mirror, the radiation pressure affects the movable mirror. In turn, the mirror affects the light intensity by controlling the cavity length and, therefore, resonance frequencies. Researchers have used this photon-phonon coupling in various experiments ranging from quantum optical measurements of mechanical motion to back-action cooling [14, 15].

Suppose we take an array of optomechanical systems by coupling many

cavities together, allowing the photon and phonons to hop (tunnel) between the neighboring sites. These arrays may be useful for quantum information processing (QIP) [16], understanding quantum many-body physics [17], and photon transport [18].

1.3 Polarons

A polaron is a quasi-particle formed when a quantum excitation (e.g., charge) moves slowly through a lattice, causing distortion or polarization, and this deformation moves along with the charge [3]. Generally, a polaron is formed whenever the energy gained by trapping the charge carriers exceeds the energy required to displace the surrounding atoms from their equilibrium sites, forming the local distortion. This distortion is quantified by the radius, as follows. A small radius with respect to the unit cell constant a of the lattice is called a *small* (or Holstein) polaron. Conversely, a *large* polaron is associated with local deformation extending beyond one unit cell ([19]). Polarons have been studied extensively in the condensed matter context and in different areas of ultracold physics dealing with ultracold atoms [20], polar molecules [9], and BECs [10].

In our case, to mimic the behavior of Holstein polaron transport, we consider optomechanical arrays as a platform. The Hamiltonian describing the optomechanical arrays resembles the Holstein Hamiltonian and can describe polaron formation and dynamics [3, 20]. In our project, we consider a single photon interacting with the phonon states, forming a polaron. Firstly, we obtain the polaron ground state and compare it with the single-site optomechanical eigenstate. For a specific range of applicability, we use this polaron ground state for simulation of transport for the two cavities, and then extend it to a larger array.

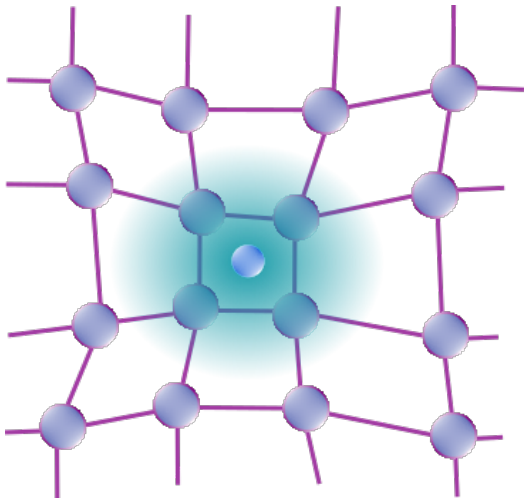

Fig. 1.2: A charge in a lattice causing distortion. This charge along with the deformation moves slowly through the lattice and called a polaron.

1.4 Outline

The main aim of this work is to investigate polaron transport in optomechanical arrays and understand which parameters the transport depends on. As a first step, we study the polaron transport in two cavities and then extend it to a one-dimensional array. The work is organized as follows: In Chapter 2, we formulate the general optomechanical Hamiltonian and Holstein Hamiltonian for polarons. The derivation of the polaron ground state via the polaron transformation is presented in Sect. 2.2.2 and 2.2.3.

Chapter 3 constitutes the central part of this work. It contains a numerical simulation of the polaron transport for arbitrary parameters. Firstly, we focus on the analysis of two cavities in section 3.2. We also derive hopping periods of polaron transport and verify them numerically in Sect 3.2.2. We confirm our results using the polaron transformation on the optomechanical Hamiltonian, as shown in appendices A.

The numerical analysis of a more complex system, namely one-dimensional array, is the subject of Chapter 4. The final Chapter 5 contains a summary of this work and a short future outlook.

2. THEORETICAL OVERVIEW

2.1 Optomechanical Hamiltonian

We start by considering a general single-cavity optomechanical set-up consisting of a single-mode resonant electromagnetic cavity whose frequency ω_{cav} depends on the position x of a mechanical resonator [14, 21]. Both the mechanical mode and the cavity mode are harmonic oscillators hence the Hamiltonian takes the form ($\hbar = 1$)

$$\hat{H}_{OM} = \omega_{cav}(\hat{x})\hat{a}^\dagger\hat{a} \ + \ \Omega_m\hat{b}^\dagger\hat{b} \ , \tag{2.1}$$

where $\hat{a}$ ($\hat{a}^\dagger$) and $\hat{b}$ ($\hat{b}^\dagger$) are the annihilation (creation) operators for the cavity mode and the mechanical mode, respectively. Ω_m is the frequency of the mechanical resonator. We are interested only in small displacements, so by the Taylor expansion of $\omega_{cav}(\hat{x})$ and keeping just the linear term, we get

$$\omega_{cav}(\hat{x}) \ \approx \ \omega_{cav} + \hat{x} \ \frac{\partial\omega_{cav}}{\partial x}.$$

Writing mechanical position in terms of creation and annihilation operators, the optomechanical Hamiltonian becomes

$$\hat{H}_{OM} = \omega_{cav}\hat{a}^\dagger\hat{a} + \Omega_m\hat{b}^\dagger\hat{b} - g\hat{a}^\dagger\hat{a}(\hat{b} + \hat{b}^\dagger) \equiv \hat{H}_o + \hat{H}_{int} \ , \tag{2.2}$$

where g is the optomechanical coupling strength. It quantifies the interaction between a single photon and a single phonon, and is given as

$$g = -\frac{\partial\omega_{cav}}{\partial x}x_{ZPF} = -\frac{\partial\omega_{cav}}{\partial x}\sqrt{\frac{\hbar}{2m\Omega_m}} \ , \tag{2.3}$$

where x_{ZPF} is the zero-point fluctuation of the Harmonic oscillator. The sign reflects that a positive displacement $x > 0$, which indicates an increase in the cavity's length leads to a decrease in its resonance frequency.

The physical interpretation of the coupling term g is that the mechanical oscillator is displaced by x_d due to radiation pressure which can be found from:

$$\frac{\hat{p}^2}{2M} + \frac{1}{2}M\Omega_m^2\hat{x}^2 - \frac{g}{x_{ZPF}}(\hat{a}^\dagger\hat{a})\hat{x} = \frac{\hat{p}^2}{2M} + \frac{1}{2}M\Omega_m^2(\hat{x} - x_d)^2$$
$$\implies x_d = \frac{2g\ x_{ZPF}}{\Omega_m}(\hat{a}^\dagger\hat{a}) \ . \tag{2.4}$$

Hence, for a single photon in the cavity, the mechanical oscillator displaces by $x_d = \frac{2g\ x_{ZPF}}{\Omega_m}$. The optomechanical interaction is cubic in nature, i.e., it contains terms like $\hat{a}_j^\dagger\hat{a}_j\hat{b}_j^\dagger$ and therefore, we can see non-linear quantum effects for the regime $g > \Omega$ [22]. From Eq. 2.4, we see that, if $g > \Omega$, then a single photon will displace the mechanical mode more than the zero-point motion. It means that, if the oscillator is in the ground state where it has a quite localized wave packet, a single photon will displace the ground state function to a distance such that a new state will be almost orthonormal to the original state.

2.1.1 Optomechanical Arrays

Now, if we consider a one-dimensional chain of optomechanical cavities and allow photon and phonon hopping in this array, the Hamiltonian with no external driving becomes [17]

$$\hat{H} = \sum_{j=1}^{n} \hat{H}_{OM,j} + \hat{H}_{hop} \ , \tag{2.5}$$

where $\hat{H}_{OM}$ is the Hamiltonian for a single-site optomechanical set-up, Eq. 2.2, furthermore the subscript j indicates the site in an optomechanical array. The hopping Hamiltonian $\hat{H}_{hop}$ is given as

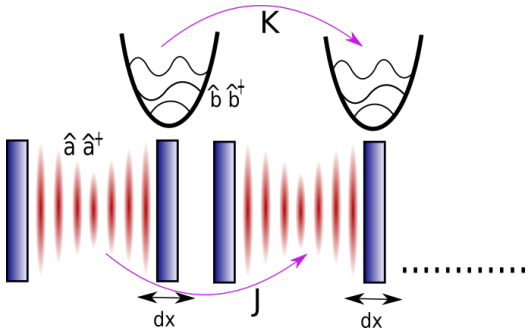

Fig. 2.1: 1-D optomechanical array with J and K as photon and phonon hopping, respectively. Operators $\hat{a}$ ($\hat{a}^\dagger$) and $\hat{b}$ ($\hat{b}^\dagger$) are the annihilation (creation) operators for the cavity mode and the mechanical mode, respectively.

$$\hat{H}_{hop} = -J \sum_{\langle i,j \rangle} (\hat{a}_i^\dagger \hat{a}_j \ + \ \hat{a}_i \hat{a}_j^\dagger) - K \sum_{\langle i,j \rangle} (\hat{b}_i^\dagger \hat{b}_j \ + \ \hat{b}_i \hat{b}_j^\dagger) \ , \tag{2.6}$$

where J and K are tunneling amplitudes of photon and phonons, respectively, and $\langle i, j \rangle$ represents nearest neighboring sites only.

2.2 Polaron Physics

2.2.1 Holstein Hamiltonian

Our main objective is the quantum simulation of the polaron dynamics, and the Holstein Hamiltonian mentioned in section 1.3, is given in the second quantization as [3, 19]:

$$\hat{H}_{Hol} = -J \sum_{\langle i,j \rangle} (\hat{c}_i^\dagger \hat{c}_j + h.c) + \Omega \sum_j \hat{b}_j^\dagger \hat{b}_j - g \sum_j \hat{c}_j^\dagger \hat{c}_j (\hat{b}_j^\dagger + \hat{b}_j) \ , \tag{2.7}$$

where $\hat{c}$ ($\hat{c}^\dagger$) and $\hat{b}$ ($\hat{b}^\dagger$) are the annihilation (creation) operators for electrons and phonons, respectively. $\langle i, j \rangle$ represents summation over nearest neighbouring sites only, and J is the nearest neighbour hopping amplitude for electron. Ω is the phonon frequency, and g is the electron-phonon cou-

pling. This Hamiltonian (Eq. 2.7) describes a *small* or *Holstein* polaron [3]; see section 1.3.

If we consider a strong electron-phonon coupling, then the hopping term can be treated as a small perturbation [19]. The last two terms of Holstein Hamiltonian can be diagonolized using the Lang-Firsov Transformation or the Polaron Transformation [19, 23].

2.2.2 Polaron Transformation

The main idea of the polaron transformation is to describe the system in the polaron frame such that the environment already dresses the system. The dressed system or polaron takes the significant effects of system-environment interaction g into consideration, and hence the hopping term can be treated as a perturbation.

On the application of the canonical transformation, the new Hamiltonian is given as,

$$\hat{H}' = e^{\hat{S}} \hat{H} e^{-\hat{S}} \ . \tag{2.8}$$

We now choose a $\hat{n} = \hat{c}^\dagger \hat{c}$ dependent displacement transformation of the mechanical resonator, $\hat{S} = -\frac{g}{\Omega} \sum_j \hat{c}_j^\dagger \hat{c}_j (\hat{b}_j^\dagger - \hat{b}_j)$. This operator $\hat{S}$ is anti-Hermitian i.e, $\hat{S}^\dagger = -\hat{S}$ and, can be expanded in the form of commutators using the identity:

$$e^{\hat{S}} \hat{O} e^{-\hat{S}} = \hat{O} + [\hat{S}, \hat{O}] + \frac{[\hat{S}, [\hat{S}, \hat{O}]]}{2!} + \dots$$

Using the above identity on annihilation and creation operators for charge and phonons as mentioned in Eq. 2.7, we obtain the transformed operators in the polaron frame as,

$$c'_j = \hat{c}_j X_j, \qquad \hat{c}'^{\dagger}_j = \hat{c}^{\dagger}_j X^{\dagger}, \tag{2.9a}$$

$$\hat{b}'_j = \hat{b}_j + \frac{g}{\Omega}\hat{c}^{\dagger}_j\hat{c}_j, \qquad \hat{b}'^{\dagger}_j = \hat{b}^{\dagger}_j + \frac{g}{\Omega}\hat{c}^{\dagger}_j\hat{c}_j, \tag{2.9b}$$

where $X_j = e^{\frac{g}{\Omega}(\hat{b}^{\dagger}_j - \hat{b}_j)}$ and the transformed Hamiltonian is finally given by,

$$\hat{H}' = \hat{H}'_o + \hat{H}'_{hop} \tag{2.10a}$$

$$\hat{H}'_o = \Omega \sum_j \hat{b}^{\dagger}_j\hat{b}_j - \frac{g^2}{\Omega}\sum_j \hat{c}^{\dagger}_j\hat{c}_j \tag{2.10b}$$

$$\hat{H}'_{hop} = -J\sum_{\langle i,j\rangle}(\hat{c}^{\dagger}_i\hat{c}_j X^{\dagger}_i X_j + h.c) \tag{2.10c}$$

Now, $\hat{H}'_o$ is diagonal, and the effect of the interaction has been reduced to just a shift of energy by $\frac{g^2}{\Omega}$, for a single electron. We can find energy eigenvalues of $\hat{H}'_o$ and treat $\hat{H}'_{hop}$ as a small perturbation.

The physical meaning of the canonical transformation is to shift the ions equilibrium position at the sites where the electron is present. As can be seen from the following transformed position operators:

$$\begin{aligned}
\langle x'_j\rangle &= \langle \hat{b}'^{\dagger}_j + \hat{b}'_j\rangle \\
&= \langle \hat{b}^{\dagger}_j + \hat{b}_j + \frac{2g}{\Omega}\hat{c}^{\dagger}_j\hat{c}_j\rangle \\
\langle x'_j\rangle &= \langle x_j\rangle + \frac{2g}{\Omega}\langle \hat{c}^{\dagger}_j\hat{c}_j\rangle
\end{aligned}$$

We see that the new equilibrium position is the sum of the old equilibrium position plus a displacement proportional to the average number of electrons present at that site with a factor $\frac{2g}{\Omega}$. This shift in the ions position resembles the optomechanical displacement calculated in Eq. 2.4.

We can read from the Hamiltonian $\hat{H}'_o$ in Eq. 2.10, the energy eigenvalues for a single electron are given as

$$E_j = -\frac{g^2}{\Omega} + n_j\Omega, \tag{2.11}$$

where n_j is the phonon number. We see that the ground state energy is given by $E_p = -\frac{g^2}{\Omega}$, also called *polaron binding energy* or lattice deformation energy gained due to a presence of an electron [19]. The transformed Hamiltonian gives an unperturbed N-degenerate ground state, and every state consists of a localized electron trapped at a particular site.

The first-order correction of $\hat{H}_{hop}$ lifts the degeneracy. Now, using the basis representation $|\, l, n_l \,\rangle = c_l^\dagger |\, n_l \,\rangle$, where l is the site occupied by the electron and $|\, n_l \,\rangle$ represents the phonon states at the site. Thus, to compute the first-order correction, we calculate $\langle\, i, (n_i = 0) \,|\hat{H}_{hop}|\, j, (n_j = 0) \,\rangle$ as:

$$\begin{aligned} J_{eff} &= J \,\langle\, 0, i \,|\hat{c}_i^\dagger \hat{c}_j X_i^\dagger X_j|\, j, 0 \,\rangle \\ &= J \,\langle\, 0, i \,|\hat{c}_i^\dagger \hat{c}_j e^{-\frac{g}{\Omega}(\hat{b}_i^\dagger - \hat{b}_i)}\; e^{\frac{g}{\Omega}(\hat{b}_j^\dagger - \hat{b}_j)}|\, j, 0 \,\rangle \\ J_{eff} &= J e^{-\frac{g^2}{\Omega^2}} \;. \end{aligned} \tag{2.12}$$

We used the Baker-Cambell-Haussdorf relation for two non-commuting operators, $e^{\alpha(A+B)} = e^{\alpha A}\; e^{\alpha B}\; e^{-\alpha^2[A,B]}$, to expand the exponentials in the second step. Hence for the energy, we have seen at the zeroth order, the lattice deformation traps the electron at a given site and the first-order correction lifts the degeneracy with an exponentially reduced nearest neighbour tunneling compared to the tunneling amplitude in old Holstein Hamiltonian, Eq. 2.7.

2.2.3 Polaron Ground State

The transformed Hamiltonian (Eq.2.10b) is diagonal, and there are N-degenerate unperturbed single electron ground states given as, $|\,GS, j\,\rangle' = \hat{c}_j'^{\dagger}|\,0\,\rangle$, where $\hat{c}_j'^{\dagger}$ gives the presence of electron at the jth site, and $|\,0\,\rangle$ is the ground state of phonon. It can be projected back into the original basis by considering the inverse transformation.

$$\begin{aligned}
\hat{c}_j'^{\dagger}|\,0\,\rangle &= e^{\hat{S}}|\,GS, j\,\rangle \\
|\,GS, j\,\rangle &= e^{-\hat{S}}\hat{c}_j'^{\dagger}|\,0\,\rangle \\
&= e^{\frac{g}{\Omega}(\hat{b}_j^{\dagger}-\hat{b}_j)}\ \hat{c}_j'^{\dagger}|\,0\,\rangle \\
&= e^{\frac{g}{\Omega}\hat{b}_j^{\dagger}}\ e^{-\frac{g}{\Omega}\hat{b}_j}\ e^{-\frac{g^2}{2\Omega^2}}\ \hat{c}_j'^{\dagger}|\,0\,\rangle \\
|\,GS, j\,\rangle &= e^{-\frac{g^2}{2\Omega^2}}\ e^{\frac{g}{\Omega}\hat{b}_j^{\dagger}}\ \hat{c}_j'^{\dagger}|\,0\,\rangle
\end{aligned} \tag{2.13}$$

We see that the phononic part of ground state turns out to be the coherent state of Harmonic oscillator in original basis. This can be interpreted as an electron dressed by a phonon cloud. Some properties of coherent state are given before proceeding further.

Coherent state of Harmonic Oscillator

A coherent state $|\,\alpha\,\rangle$ is the eigenstate of an annihilation operator $\hat{b}$ with eigenvalue α, formally given as,

$$\hat{b}|\,\alpha\,\rangle = \alpha|\,\alpha\,\rangle.$$

The coherent state can be expressed in terms of energy eigenstates as

$$|\,\alpha\,\rangle = \sum_{n=0}^{\infty}|\,n\,\rangle\langle\,n\,|\alpha\rangle = \alpha_o^2\sum_{n=0}^{\infty}\frac{(\alpha)^n}{\sqrt{n!}}|\,n\,\rangle\ ,$$

where, $|\,n\,\rangle = \frac{(\hat{b}^{\dagger})^n}{\sqrt{n!}}|\,0\,\rangle$. Using the normalization condition $|\langle\alpha|\alpha\rangle|^2 = 1$, we can find the coefficient α_o as:

$$\alpha_o^2 \sum_{n=0}^{\infty} \left| \frac{\alpha^n}{\sqrt{n!}} \right|^2 \;=\; \alpha_o^2 \; e^{|\alpha|^2} = 1 \quad \Longrightarrow \; \alpha_o = e^{-\frac{|\alpha|^2}{2}}$$

Hence, the coherent state is given as,

$$|\,\alpha\,\rangle = e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |\,n\,\rangle = e^{-\frac{|\alpha|^2}{2}} \; e^{\alpha \hat{b}_j^\dagger} |\,0\,\rangle$$

with $\alpha = \frac{g}{\Omega}$ for the polaron ground state, as we see from Eq. 2.13.

3. NUMERICAL ANALYSIS OF POLARON TRANSPORT

The previous chapter, provided the theoretical background and the Hamiltonian formulation of an optomechanical set-up and the Holstein polaron. We also saw that the polaron ground state is coherent state. The polaron transformation can also be applied to the Hamiltonian of an optomechanical array, Eq. (2.5), and the photon and phonon hopping can then be treated as a small perturbation, see appendices A. We can then perturbatively calculate the ground state energy and ground-level eigenstate.

We notice that the Hamiltonian for our optomechanical system without phonon hopping, Eq. (2.5), and the one for a Holstein polaron, Eq. (2.7), are of similar form with a constant factor of photon energy extra in the former. The photon-phonon interaction for an optomechanical Hamiltonian is given by the displacement of mirror due to radiation pressure, whereas a condensed matter Holstein polaron originates from a lattice deformation due to presence of an electron. We consider a single photon interacting with the phonon forming a polaron quasi-particle for our system. In this section, we will first calculate the polaron ground state for the single optomechanical cavity and then check the range of approximations involved for our system. Further, we will use the polaron ground state as the initial state for the simulation and see how the polaron transports to the neighbouring sites.

3.1 Polaron Ground State for Optomechanical Cavity

We can calculate the ground state and energy eigenvalue for an optomechanical cavity using the polaron transformation; see previous page for reference. We can also assume the ground-state to be a coherent state and thus use a variational approach to calculate the ground state energy eigenvalue and the coefficient α for the coherent state based on the discussion on the previous page.

Taking the many-body state as $|\,1,\alpha\,\rangle$, where 1 represents the photon number present in the cavity and $|\,\alpha\,\rangle$ as a coherent state given by,

$$|\,\alpha\,\rangle \;=\; e^{-\frac{|\alpha|^2}{2}} \sum_n \frac{\alpha^n}{\sqrt{n!}}\,|\,n\,\rangle_{phonons}\;. \tag{3.1}$$

Considering this state to be a solution of the Hamiltonian, Eq. (2.2), we can solve the following Eq. $\langle\,\alpha,1\,|\hat{H}_{OM}|\,1,\alpha\,\rangle = \langle\,\alpha,1\,|E|\,1,\alpha\,\rangle$. We get a quadratic equation in α

$$E = \Omega_m\;\alpha^2 - 2\alpha g + \omega_{cav}. \tag{3.2}$$

We consider E as a function of α and find α that minimizes the energy (minimum energy is an upper bound of the ground state energy). In this manner, we obtain α and the approximate ground state energy as:

$$\alpha = \frac{g}{\Omega_m}\;, \qquad\qquad E_{GS} = \omega_{cav} - \frac{g^2}{\Omega_m}\;. \tag{3.3}$$

To verify our solution and its range of applicability, we plot oscillator state populations ($|c_n|^2$) and energy eigenvalues of the numerically obtained ground state of the single-site optomechanical Hamiltonian, Eq. (2.2) and our analytical polaron ground state (inserting the value of α from Eq. 3.3 in Eq. 3.1) in Fig. 3.1. For the numerical calculation, we expand the quantum state $|\,\psi\,\rangle = \sum_{n=0}^{4} c_{1,n}|\,1,n\,\rangle$, where n represents the oscillator states and 1 represents a single photon. We then solve the eigenvalue problem

$\hat{H}_{OM}|\,\psi\,\rangle = E|\,\psi\,\rangle$, and calculate oscillator state populations, $|c_n|^2$.

From Fig. 3.1, we see that the coefficients of polaron ground state and ground-level eigenstate are similar for $g \lesssim \Omega_m$, but begin to deviate for $g \gtrsim \Omega_m$. Increasing Ω_m allows coefficients to remain similar for larger g, and agreement is found in the range $g \lesssim \Omega_m$. A similar trend is seen when we compare numerically obtained ground-level energy eigenstate and analytical energy ground-state, Eq. (3.3).

3.2 Analysis for Two Cavities

Now, we know the range of applicability, therefore applying it for our analysis of two cavities using a many-body photon-phonon basis and polaron basis. From Fig. 3.1, we see that the polaron ground state is similar to the ground-level energy eigenstate for $g \lesssim \Omega$. This result agrees with the non-linear effects that one can get for $g > \Omega$ as discussed in section 2.1, Eq. (2.4).

We consider a single photon that can tunnel between two cavities and interact with the mechanical resonators. The Hamiltonian for such a system is given by,

$$\begin{aligned}\hat{H} = \sum_{j=1}^{2} &[\omega_j \hat{a}_j^\dagger \hat{a}_j + \Omega_j \hat{b}_j^\dagger \hat{b}_j - g\ \hat{a}_j^\dagger \hat{a}_j (\hat{b}_j^\dagger + \hat{b}_j)] \\ &- J(\hat{a}_1^\dagger \hat{a}_2 + \hat{a}_1 \hat{a}_2^\dagger) - K(\hat{b}_1^\dagger \hat{b}_2 + \hat{b}_1 \hat{b}_2^\dagger), \end{aligned} \tag{3.4}$$

where ω_j and Ω_j is the resonant photon and phonon frequency, respectively, for the jth cavity. g is the single photon-phonon coupling, and J and K are photon and phonon tunneling amplitudes, respectively.

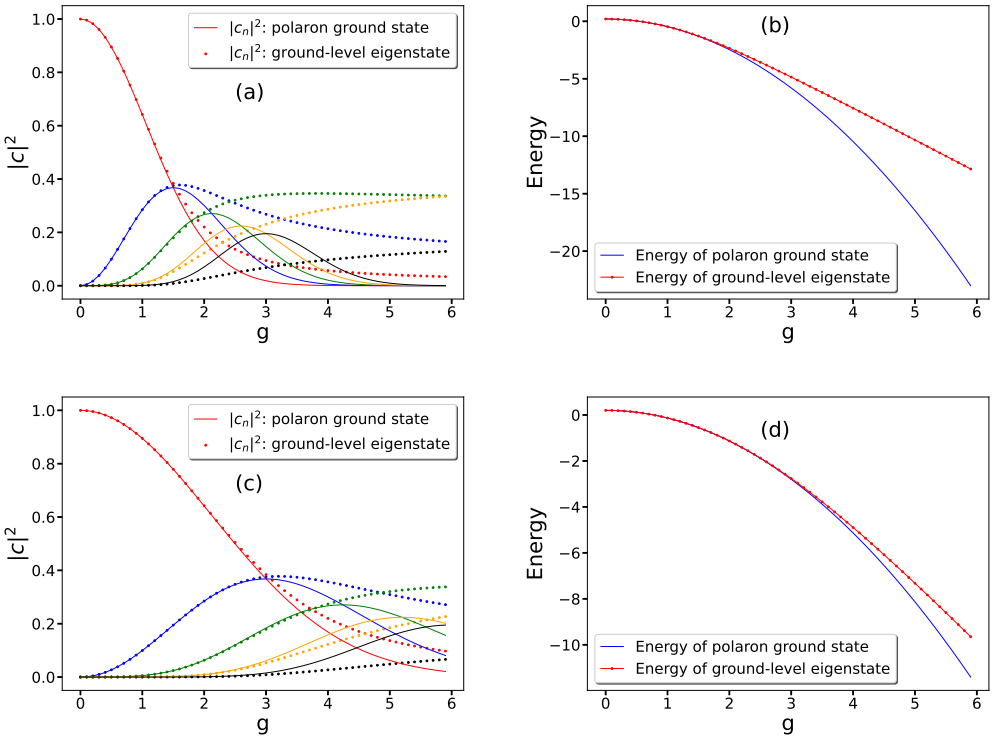

Fig. 3.1: Comparison of occupations ($P_n = |c_n|^2$, $n = [0, 4]$) and energy eigenvalues in the ground-level eigenstate of single-site optomechanical Hamiltonian calculated numerically (dotted lines) solving eigenvalue problem $\hat{H}_{OM}|\,\psi\,\rangle = E|\,\psi\,\rangle$ with $|\,\psi\,\rangle = \sum_{n=0}^{4} c_{1,n}|\,1, n\,\rangle$, and our polaron ground state calculated analytically (solid lines) in Eq. 3.1 and Eq. 3.3 for various coupling strengths g. The oscillator frequency Ω_m is 1.5 in (a) and (b) and, 3.0 in (c) and (d). Oscillator states n ranges from 0 (red) to 4 $(black)$ in (a) and (c). Occupations are similar for $g \lesssim \Omega_m$ and depend on the frequency Ω_m. As we increase Ω_m, occupations are similar for larger range of couplings g. Energy eigenvalues also follow the same trend. $\omega_{cav} = 0.2$ in all the cases.

3.2.1 Many-body photon-phonon basis

Let us now take the general wavefunction for this system as a tensor product of the photon and the phonon states,

$$|\,\Psi\,\rangle = \sum_{m_1,m_2=0}^{4} c_{1,m_1,m_2}|\,1,m_1,m_2\,\rangle + c_{2,m_1,m_2}|\,2,m_1,m_2\,\rangle\ , \tag{3.5}$$

where 1 is the state with a photon present in the left cavity and 2 in the right cavity. m_1 and m_2 are the quantum numbers of phonon vibrations of the cavity one and two, respectively. c_{j,m_1,m_2} for $j = 1, 2$ are the time-dependent coefficients to be determined.
Proceeding to the time-dependent Schrödinger's equation for the Hamiltonian 3.4, we find the equation of motion for two cavities as,

$$\begin{aligned} i\ \dot{c}_{1,m_1,m_2} &= \omega_1\ c_{1,m_1,m_2} + (\Omega_1\ m_1 + \Omega_2\ m_2)\ c_{1,m_1,m_2} \\ &- g\left(\sqrt{m_1+1}\ c_{1,m_1+1,m_2} + \sqrt{m_1}\ c_{1,m_1-1,m_2}\right) - J\ c_{2,m_1,m_2} \\ &- K\left[\sqrt{(m_1+1)\ m_2}\ c_{1,m_1+1,m_2-1} + \sqrt{m_1(m_2+1)}\ c_{1,m_1-1,m_2+1}\right], \end{aligned} \tag{3.6a}$$

$$\begin{aligned} i\ \dot{c}_{2,m_1,m_2} &= \omega_2\ c_{2,m_1,m_2} + (\Omega_1\ m_1 + \Omega_2\ m_2)\, c_{2,m_1,m_2} \\ &- g\left(\sqrt{m_2+1}\ c_{2,m_1,m_2+1} + \sqrt{m_2}\ c_{2,m_1,m_2-1}\right)\ - J\ c_{1,m_1,m_2} \\ &- K\left[\sqrt{(m_1+1)\ m_2}\ c_{2,m_1+1,m_2-1} + \sqrt{m_1\ (m_2+1)}\ c_{2,m_1-1,m_2+1}\right]. \end{aligned} \tag{3.6b}$$

We now want to separately plot the photon and the phonon populations for two coupled cavities. Therefore, we calculate the reduced density matrix for the photon and phonons in each cavity separately, and the diagonal elements of which give us populations $P_{photon/phonon_j}$, for the photon and phonons in j^{th} cavity. $c_{j,m_1,m2}$ are the time-dependent coefficients that we find after solving coupled equations of motion, Eq. 3.6a and Eq. 3.6b.

$$P_{photon_1} = \sum_{m_1'',m_2''=0}^{4} |c_{1,m_1'',m_2''}|^2 \ , \tag{3.7a}$$

$$P_{photon_2} = \sum_{m_1'',m_2''=0}^{4} |c_{2,m_1'',m_2''}|^2 \ , \tag{3.7b}$$

$$P_{phonon_1} = \sum_{m_2''=0}^{4} (|c_{1,m_1,m_2''}|^2 + |c_{2,m_1,m_2''}|^2) \ , \tag{3.7c}$$

$$P_{phonon_2} = \sum_{m_1''=0}^{4} (|c_{1,m_1'',m_2}|^2 + |c_{2,m_1'',m_2}|^2) \ . \tag{3.7d}$$

For a numerical solution, we used the XMDS (Extensible Multi-Dimensional Simulator) [24]. The XMDS code used the ARK89 algorithm. Initially, a photon is placed in the first cavity and interacts with the phonon modes forming a polaron. Hence as an initial state we take a tensor product of a polaron ground state $|\,\alpha\,\rangle$ in the first cavity and $|\,0\,\rangle$ (ground state of harmonic oscillator, i.e., phonon ground state) in the second cavity.
The initial condition is thus, $c_{1,m_1,m_2=0} = e^{-\frac{|\alpha|^2}{2}} \sum_{m_1=0}^{4} \frac{\alpha^{m_1}}{\sqrt{m_1!}}$ for $\alpha = \frac{g}{\Omega_m}$ and $c_{2,m_1,m_2=0} = 0$. In simulations, we consider both the photon and phonon hopping. In Fig. 3.2, we see the photon initially in the cavity one couples with phonon modes and forms a polaron and tunnels periodically between the two cavities.

3.2.2 Polaron Basis

Now, we want to analyze for the two cavities from polaron's frame of reference. We assume that a single photon present in the jth cavity interacts with the phonon and forms a polaron. In polaron's frame of reference, this photon-phonon interaction is present in the jth cavity, and the rest of the cavities are empty. Hence, the wavefunction is given as a superposition of polaron present in each cavity.

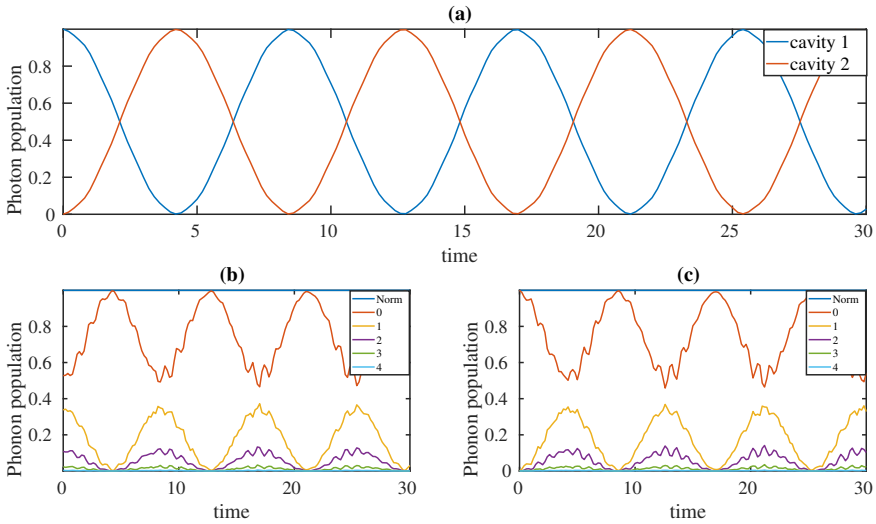

Fig. 3.2: Photon and phonon population for two cavities with $\omega_1 = \omega_2 = \pi$, $\Omega_1 = \Omega_2 = 2.0\pi$, $g = 1.6\pi$, $J = 0.2\pi$, $K = 0.2\pi$ and $n_{phonons} = 5$. In (a), the photon population in cavity one (blue) and cavity two (orange) exhibits Rabi oscillations. Panel (b) and (c) show the phonon population of the ground state (orange) to the fourth excited state (blue) in cavity one and cavity two, respectively. The Norm is conserved in both cases. The photon interacts with the phonon modes and forms a polaron, and this polaron tunnels periodically to another cavity.

$$|\,\Psi'(t)\,\rangle = c_1(t)|\,\psi'_{pol1}\,\rangle + c_2(t)|\,\psi'_{pol2}\,\rangle\ , \tag{3.8}$$

where $|\,\psi'_{pol1}\,\rangle = \sum_{m'_1=0}^{4}\ d_{1,m'_1,0}|\,1, m'_1, 0\,\rangle$ represents a polaron present in cavity one and $|\,\psi'_{pol2}\,\rangle = \sum_{m'_2=0}^{4}\ d_{2,0,m'_2}|\,2, 0, m'_2\,\rangle$ represents a polaron present in cavity two. d_{m_i} is the coefficient of polaron ground state, given as

$$d_{m_i} = e^{-\frac{|\alpha_i|^2}{2}}\,\frac{\alpha_i^{m_i}}{\sqrt{m_i!}}\ , \tag{3.9}$$

where $\alpha_i = \frac{g}{\Omega_i}$, and $c_i(t)$ are the two-mode coefficients to be determined.

Using Schrödinger's equation ($\hbar = 1$), we insert the ansatz (3.8) into $i\frac{d}{dt}\ |\,\Psi'(t)\,\rangle = H|\,\Psi'(t)\,\rangle$, and to obtain the time evolution of both the co-

efficients separately, project the wavefunction as in $\langle\, \psi_{pol1} \mid\, i\frac{d}{dt}\, \mid \Psi'(t)\,\rangle = \langle\, \psi_{pol1} \mid\, H \mid \Psi'(t)\,\rangle$:

$$
\begin{aligned}
i\dot{c}_1(t) = c_1(t)\Bigg[\omega_1 + \Omega_1 \left(\sum_{m_1'=0}^{4} m_1' |d_{1,m_1,0}|^2\right) - g \sum_{m_1',m_1=0}^{4} \Bigg(\sqrt{m_1'+1}\; d_{1,m_1',0}\; d^*_{1,m_1,0}\; \delta_{m_1,m_1'+1} \\
+ \sqrt{m_1'}\; d_{1,m_1',0}\; d^*_{1,m_1,0}\; \delta_{m_1,m_1'-1}\Bigg) + K \left(\sum_{m_1',m_1=0}^{4} \sqrt{m_1'}\; d_{1,m_1',0}\; d^*_{1,m_1,0}\; \delta_{m_1,m_1'-1}\; \delta_{0,1}\right)\Bigg] \\
- J\; c_2(t) \left(\sum_{m_2',m_1=0}^{4} d_{2,0,m_2'}\; d^*_{1,m_1,0}\; \delta_{m_1,0}\; \delta_{0,m_2'}\right),
\end{aligned}
$$

and for the second coefficient, we project, $\langle\, \psi_{pol2} \mid i\frac{d}{dt} \mid \Psi'(t)\,\rangle = \langle\, \psi_{pol2} \mid H \mid \Psi'(t)\,\rangle$,

$$
\begin{aligned}
i\dot{c}_2(t) = c_2(t)\Bigg[\omega_2 + \;\Omega_2 \left(\sum_{m_2'=0}^{4} m_2' |d_{2,0,m_2}|^2\right) - g \sum_{m_2',m_2=0}^{4} \Bigg(\sqrt{m_2'+1}\; d_{2,0,m_2'}\; d^*_{2,0,m_2}\; \delta_{m_2,m_2'+1} \\
+ \sqrt{m_2'}\; d_{2,0,m_2'}\; d^*_{2,0,m_2}\; \delta_{m_2,m_2'-1}\Bigg) + K \left(\sum_{m_2',m_2=0}^{4} \sqrt{m_2'}\; d_{2,0,m_2'}\; d^*_{2,0,m_2}\; \delta_{m_2,m_2'-1}\; \delta_{0,1}\right)\Bigg] \\
- J\; c_1(t) \left(\sum_{m_1,m_2'=0}^{4} d_{1,m_1,0}\; d^*_{2,0,m_2}\; \delta_{m_1',0}\; \delta_{0,m_2'}\right).
\end{aligned}
$$

The equations above can be written in simpler form as,

$$i\dot{c}_1(t) = c_1(t)[\omega_1 + \Omega_{1_{eff}} - g_{1_{eff}}] - c_2(t)\; J_{1_{eff}}\;, \tag{3.10a}$$

$$i\dot{c}_2(t) = c_2(t)[\omega_2 + \Omega_{2_{eff}} - g_{2_{eff}}] - c_1(t)\; J_{2_{eff}}\;, \tag{3.10b}$$

where,

$$\Omega_{1_{eff}} = \Omega_1 \left(\sum_{m_1=0}^{4} m_1 |d_{1,m_1,0}|^2 \right) , \tag{3.11a}$$

$$\Omega_{2_{eff}} = \Omega_2 \left(\sum_{m_2=0}^{4} m_2 |d_{2,0,m_2}|^2 \right) , \tag{3.11b}$$

$$g_{1_{eff}} = g \sum_{m_1=0}^{4} \left(\sqrt{m_1+1}\ d_{1,m_1+1,0}\ d^*_{1,m_1,0}\ + \sqrt{m_1}\ d_{1,m_1-1,0}\ d^*_{1,m_1,0} \right) , \tag{3.11c}$$

$$g_{2_{eff}} = g \sum_{m_2=0}^{4} \left(\sqrt{m_2+1}\ d_{2,0,m_2+1}\ d^*_{2,0,m_2}\ + \sqrt{m_2}\ d_{2,0,m_2-1}\ d^*_{2,0,m_1} \right) , \tag{3.11d}$$

$$J_{1_{eff}} = J(d_{2,0,0}\ d^*_{1,0,0}) = J \left(e^{-\frac{|\alpha_2|^2}{2}} e^{-\frac{|\alpha_1|^2}{2}} \right) \tag{3.11e}$$

$$J_{1_{eff}} = J(d_{1,0,0}\ d^*_{2,0,0}) = J \left(e^{-\frac{|\alpha_1|^2}{2}} e^{-\frac{|\alpha_2|^2}{2}} \right) \tag{3.11f}$$

Now to further simplify the equations and solve the coupled differential equations we take $\Omega_1 = \Omega_2 = \Omega$, this implies our $\alpha = \frac{g}{\Omega}$, is same for both cavities and hence, coefficients d_{m_i} as in Eq. 3.9 are also the same. Furthermore, we can now take all the effective parameters to be equal for each cavity, i.e., $\Omega_{1_{eff}} = \Omega_{2_{eff}}$, $g_{1_{eff}} = g_{2_{eff}}$ and, $J_{1_{eff}} = J_{2_{eff}} = Je^{-|\frac{g}{\Omega}|^2}$.

The two coupled differential equations (Eq. 3.10) with the assumptions mentioned in the previous paragraph, can now be solved for the initial conditions $c_1(0) = 1, c_2(0) = 0$, and we obtain,

$$c_1(t) = e^{i(-\omega-\Omega_{eff}+g_{eff})t}\ \cos(J_{eff}t)\ , \tag{3.12a}$$

$$c_2(t) = i\ e^{i(-\omega-\Omega_{eff}+g_{eff})t}\ \sin(J_{eff}t)\ . \tag{3.12b}$$

We see that the phonon tunneling term does not appear as it would cause dynamics outside the ansatz (3.8). The photon tunneling, J_{eff} turns out to

be exponentially reduced, as we had seen earlier in section 2.2. We get the same results applying the polaron transformation to the optomechanical Hamiltonian; see appendices A.

We can calculate polaron populations in each cavity using the time-dependent coefficients obtained in Eq. 3.12,

$$|c_1(t)|^2 = |\cos(J_{eff}t)|^2 = \left|\frac{1+\cos(2J_{eff}t)}{2}\right| , \tag{3.13a}$$

$$|c_2(t)|^2 = |\sin(J_{eff}t)|^2 = \left|\frac{1-\cos(2J_{eff}t)}{2}\right| . \tag{3.13b}$$

Hence, the analytical formula for the Rabi Period of polaron transport is given as,

$$T = \frac{2\pi}{2J_{eff}} = \frac{\pi\ e^{|\frac{g}{\Omega}|^2}}{J} . \tag{3.14}$$

From the analytical formula, we see the T is inversely proportional to photon tunneling amplitude J. Thus if we keep everything constant and increase the value of J, the polaron tunneling period decreases. Furthermore, there is an exponential increase in the time period if we increase photon-phonon coupling g , and it decreases exponentially on increasing phonon's frequency Ω.

Let us now compare the plots of the analytical polaron period, and the numerical T versus g, Ω, J, and K. Numerical period is calculated from photon tunneling plots using the many-body photon-phonon basis. In Fig. 3.3, we can see that the analytical T resembles the numerical results. In (a), it increases exponentially as photon-phonon coupling g becomes large and, decreases on increasing phonon frequency Ω as seen in (b). This happens because as we increase the g, the photon interacts more with the phonons of that cavity and takes more time to hop. On increasing the Ω, transition

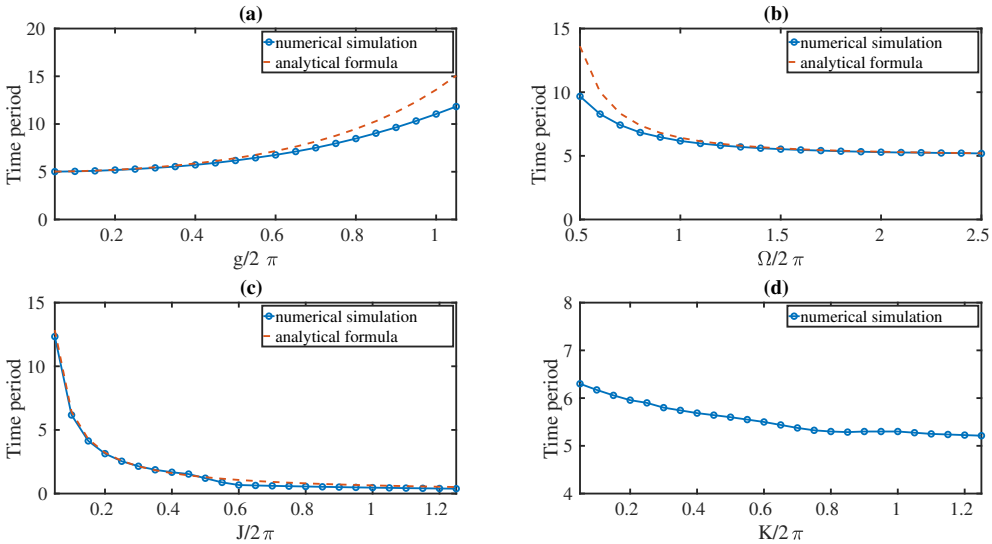

Fig. 3.3: Comparison of the analytical formula of T (dotted orange lines) as in Eq. 3.14 and the numerical Time Period (solid blue lines) calculated from photon tunneling plots for various g in (a), Ω in (b), and J in (c). The analytical and numerical polaron period agree for a large range of g, Ω, and J. We only plot the numerical T vs K in (d) as there is no dependence of K in the analytical formula. The Time Period is not affected much on changing the phonon tunneling amplitude. The constant values for each case are given as: $\omega = \pi,\ \Omega = 2\pi,\ g = \pi,\ J = 0.2\pi$ and $K = 0.2\pi$

between phonon states become fast, and it takes less time to couple with photon, resulting in a decrease in the Rabi Period. In (c), we can see inverse T proportionality with photon tunneling amplitude J clearly. In polaron's frame of reference, the phonon tunneling K vanishes; therefore we did not get any dependence of K in the analytical formula. Hence, we plot only numerical Time Period dependence on K in (d). We see that changing K does not affect it significantly.

Even though, phonon tunneling amplitude K does not affect the polaron's period, it does affect the phonon environment significantly, which affects the coherent polaron transport. In Fig 3.4, we keep everything the same as in Fig 3.2 but increase the phonon tunneling amplitude K and see that the polaron does not transport entirely to the other cavity periodically, like it did in the

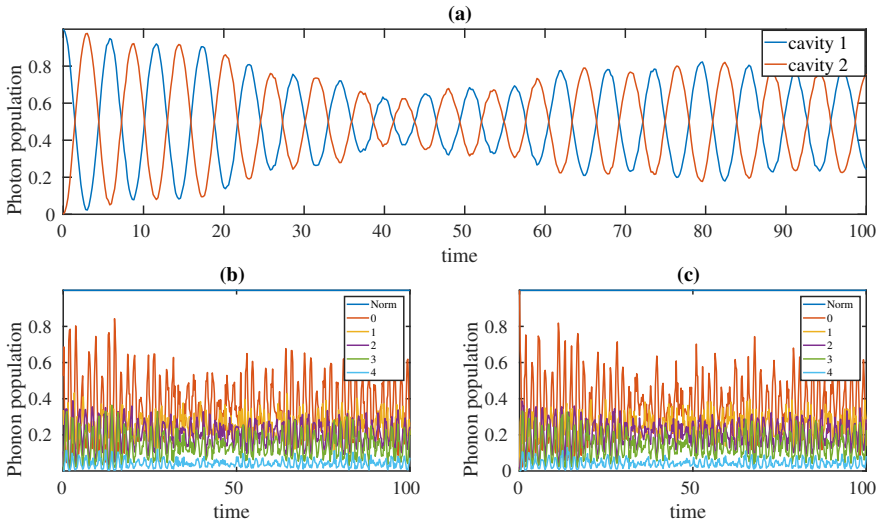

Fig. 3.4: Photon and phonon population for two cavities with $\omega_1 = \omega_2 = \pi$, $\Omega_1 = \Omega_2 = 2.0\pi$, $g = 1.0\pi$, $J = 0.2\pi$, $K = 1.6\pi$ *and* $n_{phonons} = 5$. As we increase the K from 0.2π in Fig 3.2 to 1.6π, we observe the dynamics become more complicated than before, and photon does not transport entirely to the other cavity, and polaron formation is hindered.

previous case with small K.

This chapter, described our numerical analysis of polaron transport for two cavities. We also calculated the analytical formula for the polaron tunneling period and compared it with the numerical simulations. We studied the parameters that affected the polaron transport and discussed how the phonon environment could affect the system's dynamics. Now, we will extend the dynamics to a 1-D optomechanical array.

4. EXTENSION TO 1-D OPTOMECHANICAL ARRAY

In this chapter, let us now consider a larger 1-D optomechanical array, and we will numerically simulate the polaron transport in that array. We begin by considering seven cavities and a single photon present in a cavity, and the rest of the cavities are in phonon ground state like in the two cavities case. We will first use the many-body photon-phonon basis for this case, and after that, we will increase our array to hundred cavities. Using the many-body basis becomes computationally challenging as the dimension of Hilbert space given by $N_{cavity} \cdot m^{N_{cavity}}$, increases. For seven cavities and five vibrational phonon states, the dimension becomes $7 \cdot 5^7 = 546875$. Therefore, we use the polaron basis to extend the simulation for hundred cavities.

4.1 Seven Cavities

The Hamiltonian for the optomechanical array is given by Eq. 2.5; for seven cavities we take the summation up to $j = 7$. As done in section 3.2.1, we take the general state as a tensor product of photon in j^{th} cavity ($|\, 00..j..0 \,\rangle$) and phonon vibrational states in each cavity ($|\, m_1, m_2, .., m_j, .., m_7 \,\rangle$), where $m_j = [0, 2]$. Hence the many-body wavefunction turns out to be,

$$|\, \Psi \,\rangle = \sum_{m_1, m_2, .., m_7 = 0}^{2} \sum_{j=1}^{7} c_{j, m_1, m_2, ..., m_7} |\, j, m_1, m_2, ..., m_7 \,\rangle \ . \tag{4.1}$$

Now, solving the Schrödinger's equation yields a system of coupled differential equations for the coefficients,

$$
\begin{aligned}
i\ \dot{c}_{j,m_1,m_2,..,m_7} = \omega_j\ c_{j,m_1,m_2,..,m_7} + \left(\sum_{j=1}^{7} \Omega_j\ m_j\right) c_{j,m_1,m_2,..,m_7} \\
- g\left(\sqrt{m_j+1}\ c_{j,m_1,..,m_j+1,..,m_7} + \sqrt{m_j}\ c_{j,m_1,..,m_j-1,..m_7}\right) \\
- J\left(c_{j-1,m_1,m_2,..,m_7} + c_{j+1,m_1,m_2,..,m_7}\right) - K \sum_{\langle i,j\rangle} \Bigg[\sqrt{(m_i+1)\ m_j} \\
c_{j,m_1,..,m_i+1,m_j-1,..m_7} + \sqrt{m_i(m_j+1)}\ c_{j,m_1,..,m_i-1,m_j+1,..m_7}\Bigg]
\end{aligned}
\tag{4.2}
$$

Note: For photon hopping J, boundary terms are zero i.e, for $j = 0$ we have $c_{j-1,m_1,..m_7} = 0$ and for $j = 7$ we have $c_{j+1,m_1,..,m_7} = 0$ and, for phonon hopping K, summation is for every nearest neighbour.

Photon and phonon populations are given by

$$
P_{photon} = \sum_{m_1'',..,m_7''=0}^{2} |c_{j,m_1'',..,m_7''}|^2 \tag{4.3a}
$$

$$
P_{phonon} = \sum_{m_1'',..,m_7''=0}^{2} \left| \sum_{j=1}^{n} c_{j,m_1'',..,m_j'',..,m_7''} \right|^2 \tag{4.3b}
$$

We now want to plot the polaron dynamics in the seven cavity array. We start by initializing a tensor product state of a polaron ground state i.e., a coherent state in the fourth cavity and phonon ground state in the rest of the cavities. To see the dynamics of this initial state, we let the system evolve according to Eq. 4.1 and plot the photon population profile. The coherent state or the polaron can be localized on a specific cavity or delocalized on the neighbouring cavities. This spreading or non-spreading behaviour of the polaron can be calculated using the time-dependent density profile $\rho_j(t) =$

$\langle\, \Psi(t)\, |\hat{a}_j^\dagger \hat{a}_j|\, \Psi(t)\, \rangle = \sum_{m_1'',..,m_7''=0}^{2} |c_{j,m_1'',..,m_7''}|^2$ or photon population. Temporal spreading of the initial state is given by an effective width of the spatial density profile, given as,

$$\sigma(t) = \left(N \left[\sum_{j=1}^{N} \rho_j^2(t) \right] \right)^{-1}, \tag{4.4}$$

where N is the number of cavities. $\sigma(t)$ takes the value $1/N$ for a polaron to be localized at exactly one cavity and 1 to be fully delocalized [20]. Thus, by analyzing the time-dependence of $\sigma(t)$ we can determine whether the polaron remains localized or whether it spreads across the optomechanical array.

In Fig. 4.1, we plot the photon population profile and the spatial density profile for different values of g and K. We see that polaron is exactly localized initially at the fourth cavity and then delocalize to the other cavities slowly. For smaller g and K, the polaron oscillates periodically and transports coherently. As we increase g, the transport period increases as well. Moreover, on increasing K, the polaron delocalizes entirely to all the cavities. Here also, we witness that the g affects the tunneling period, and the K affects the phonon environment, which in turn breaks the polaron and photon transport looks decohered.

4.2 Hundred Cavities

Let us now extend the analysis to the hundred cavities using the polaron basis as the many-body basis becomes computationally intractable. In polaron basis, dynamics is determined by a system of coupled differential equations, and transport only depends upon effective photon tunneling amplitude J_{eff} which in turn depends upon g, Ω, and J, see Eq. 3.11). We start by keeping the initial polaron ground state in the middle cavity like previous cases and letting the system evolve. As in Fig. 4.1, we plot the photon population and the spatial density profiles to witness the polaron dynamics. In Fig 4.2, we see that on increasing g, the J_{eff} decreases, and hence the polaron does not

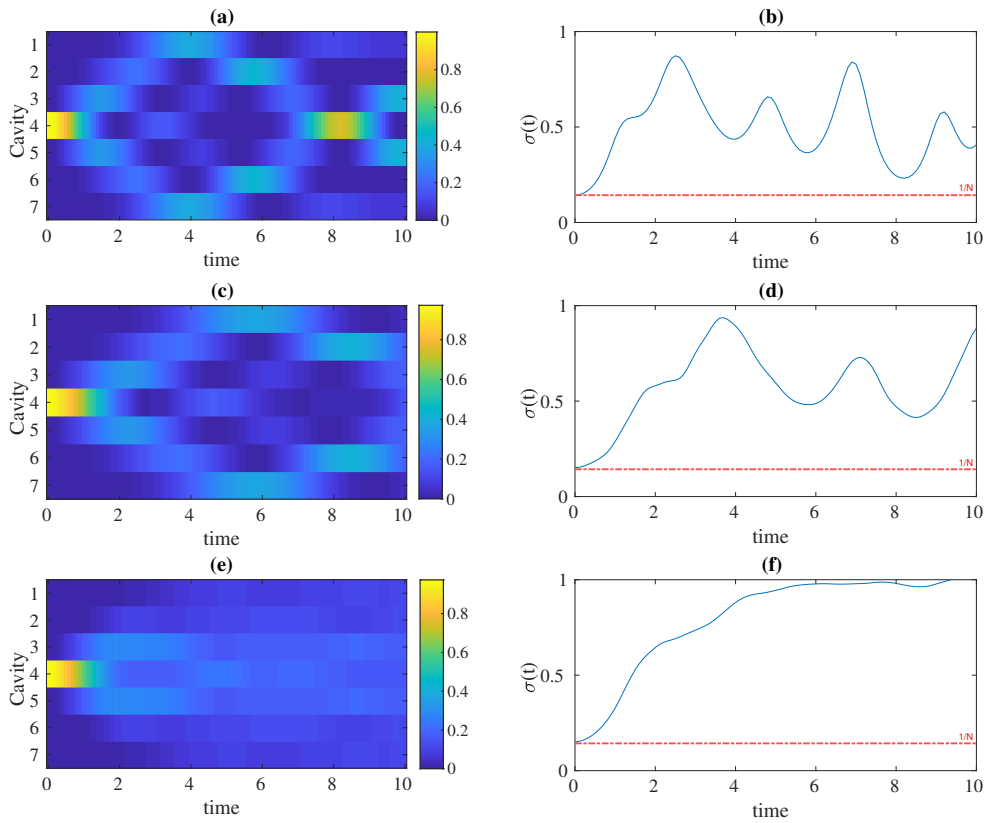

Fig. 4.1: Photon population (left panels) with colour bar indicating photon population (Eq. 4.3) and spatial density profile (right panels) with $\sigma(t)$ (Eq. 4.4) plotted as blue solid line for seven cavities. For (a) and (b), $g = 0.4\pi$, $K = 0.2\pi$ and, for (c) and (d), $g = 1.6\pi$, $K = 0.2\pi$ and, $g = 1.6\pi$, $K = 1.6\pi$ for (e) and (f). The red dotted line shows $1/N_{cavity} = 0.143$ in spatial density profiles. The constant values for each case are given as: $\omega = \pi$, $\Omega = 2\pi$, $J = 0.2\pi$. We see that the polaron transport periodically for small g and K, delocalizes to other cavities and localize to the initial cavity. The tunneling period increases as we increase the g. Moreover, on increasing K in (e), the photon delocalizes completely to other cavities and polaron breaks.

delocalize to other cavities as it did for the smaller g value. We can change the value of Ω and J similarly and see the change in transport dynamics according to the change in J_{eff}.

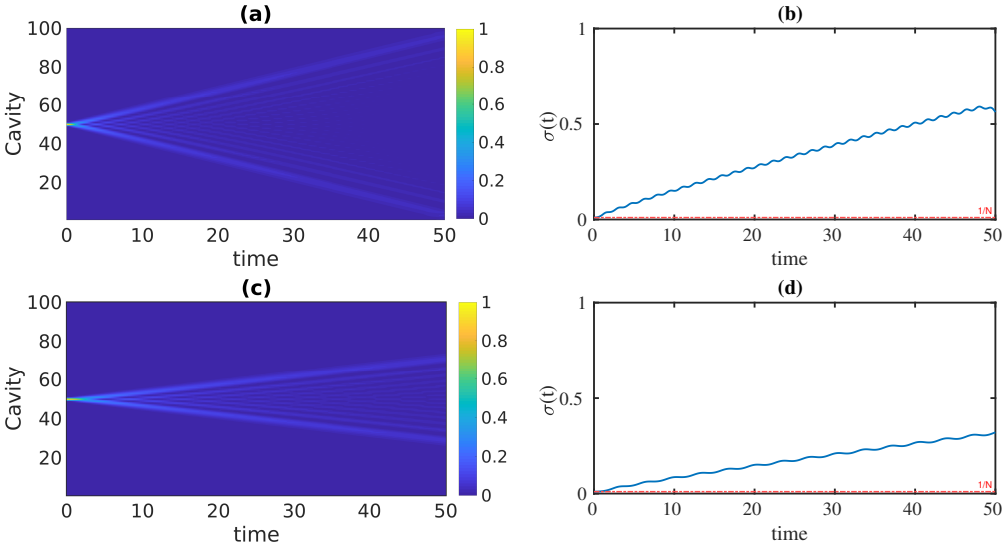

Fig. 4.2: Photon population (left panels) with colour bar indicating photon population (Eq. 3.13) and spatial density profile (right panels) with $\sigma(t)$ (Eq. 4.4) plotted as blue solid line for 100 cavities. $g = 3\pi$ in (a) and (b) and, $g = 6\pi$ in (c) and (d). The constant values for each case are given as: $\omega = \pi$, $\Omega = 6\pi$, $J = 0.2\pi$ and red dotted line shows $1/N_{cavity} = 0.01$ in spatial density profiles. The polaron is localized more in the middle and does not delocalizes to other cavities as we increase the photon-phonon coupling g.

In this chapter, we analyzed the polaron transport for a larger 1-D optomechanical array using the many-body basis and polaron. With the help of the spatial density profile, we saw that the polaron delocalizes to the other cavities periodically. The parameters g, Ω and J affects the polaron tunneling period significantly. Furthermore, the phonon tunneling K can affect the coherent transport altogether. Dynamics in the polaron basis give us the idea about the dependance of the parameters on the tunneling period, but it does not tell us about the effect of phonon environment on the dynamics. The advantage of using the many-body photon-phonon basis is that we can see the environmental effects on the dynamics.

5. CONCLUSION AND OUTLOOK

The objective of this thesis was to simulate the polaron transport on the optomechanical arrays and understand the effects of various parameters on the polaron dynamics. Initially, we studied the Hamiltonian formalism of the optomechanical arrays and compared it to the Holstein model describing polaron dynamics. Using a canonical transformation on the Holstein Hamiltonian, we obtained the polaron ground state as a coherent state. Further, we calculated the ground-level eigenstate of a single-cavity optomechanical system via a variational approach. Here, we mainly look at the resemblance of both the states and check the range of applicability for our simulation. For the numerical analysis, we work in the $g \lesssim \Omega$ range; as for the $g \gtrsim \Omega$, we can get non-linear effects. After having the limits for our work, we look at the two-cavity cases with the arbitrary parameters to see the polaron transport. Furthermore, we obtain the analytical polaron tunneling period in the polaron's frame of reference and confirm our results using the polaron transformation on the optomechanical arrays. The analytical Time Period dependence on the photon-phonon coupling g, the oscillator frequency Ω, and the photon tunneling J was compared with the numerical results. We see that they agree on a large range of parameters. We further look at the effects of the phonon tunneling amplitude on the transport and how it breaks the polaron for a larger value of K. At the end, we extend the simulation for a larger 1-D array and see the effect on delocalization of polaron to the other cavities due to g and K.

The polaron tunneling period T, depends on the three parameters, i.e. g, Ω, and J. We observe that on increasing the g, the Rabi period increases exponentially, which is expected because with the increase in photon-phonon

coupling the photon interacts more with the phonon and takes more time to tunnel to the other cavity. The reverse of this happens on increasing Ω, T decreases exponentially as the transitions between phonon states become fast, and it gets less time to couple with the photon. In the case of the third parameter, we can easily see that the J is inversely proportional with the polaron period, and hence, T reduces on increasing photon tunneling. We also witnessed that the phonon tunneling does not effect the period much.

We mainly used the two formalisms to study the polaron transport: solving the time-dependent Schrödinger's equation using a many-body basis and polaron's basis, and applying the polaron transformation on the Hamiltonian for the optomechanical arrays. The main reason for applying the transformation is to diagonolize the single-cavity Hamiltonian and treat the hopping term as a small perturbation. We obtained the same results after solving Schrödinger's equation in the polaron's frame and using the polaron transformation on the Hamiltonian. We see that the phonon tunneling K vanishes, and we get an exponentially reduced photon tunneling J_{eff} as compared to the standard J. Even though both of these methods give us an idea about the polaron dynamics and its tunneling period, they do not tell us about the effect of the phonon environment on the dynamics. We saw that, using the many-body basis, the periodic polaron transport breaks and photon looks decohered on increasing the value of K.

So far, we have not looked at the external environment's effects on the polaron transport. We can consider that both the photon and phonon modes are coupled to a bath and there is loss in mechanical and photonic mode. We can then solve Lindbladian dynamics for this system and study the effect of decoherence on the polaron transport. The other possible extension of this project could be to treat the current problem as the open quantum system (OQS). We can assume the photon mode and photon tunneling as a system and collective phonon modes of all the cavities and phonon tunneling as an environment. The local photon-phonon coupling can then be treated as the system-environment interaction. We can compute the normal modes of the

environment and use them to calculate further the correlation function between two different cavities. This correlation function can give us a measure of how strongly the environmental operator $\hat{E}_j(t)$ of the jth cavity at time t is correlated with another such operator $\hat{E}_i(t')$ of the ith cavity at time t'. The correlation function can help us determine whether an OQS is Markovian or non-Markovian i.e, whether the system loses the information to the environment or gains some back from the environment. Finally, this OQS problem can then be dealt with using numerical methods like NMQSD [25].

Appendices

Appendix A

POLARON TRANSFORMATION FOR OPTOMECHANICAL HAMILTONIAN

Hamiltonian for optomechanical arrays is given by

$$\hat{H} \;=\; \hat{H}_{OM} \;+\; \hat{H}_{hop} \tag{A.1}$$

where,

$$\hat{H}_{OM} \;=\; \sum_{j=1}^{n} (\omega_j \hat{a}_j^\dagger \hat{a}_j \;+\; \Omega_j \hat{b}_j^\dagger \hat{b}_j \;-\; g\; \hat{a}_j^\dagger \hat{a}_j (\hat{b}_j^\dagger \;+\; \hat{b}_j) \tag{A.2}$$

and hopping Hamiltonian for nearest neighbouring sites with J as photon hopping amplitude and K as phonon hopping amplitude is given by,

$$\hat{H}_{hop} \;=\; -\; J \sum_{\langle i,j \rangle} (\hat{a}_i^\dagger \hat{a}_j \;+ h.c.) \;-\; K \sum_{\langle i,j \rangle} (\hat{b}_i^\dagger \hat{b}_j \;+ h.c.) \tag{A.3}$$

Now, applying the same canonical transformation on Hamiltonian for optomechanical arrays [26], as we did for Holstein Hamiltonian in section 2.2.2.

$$\hat{H}' = \hat{H}'_{OM} + \hat{H}'_{hop} = e^S \hat{H}_{OM} e^{-S} + e^S \hat{H}_{hop} e^{-S} \tag{A.4}$$

$$\hat{H}'_{OM} = \sum_{j=1}^{n} \left(\omega_j \hat{n}_j + \Omega_j \hat{b}_j^\dagger \hat{b}_j - \frac{g^2}{\Omega} \, (\hat{n}_j)^2 \right), \tag{A.5a}$$

$$\hat{H}'_{hop} = -J \sum_{\langle i,j \rangle} \left(\hat{a}_i^\dagger \hat{a}_j X_i^\dagger X_j + h.c \right) - K \sum_{\langle i,j \rangle} \left(\hat{b}_i^\dagger \hat{b}_j + \frac{g^2}{\Omega^2} (\hat{n}_i \, \hat{n}_j) \right.$$
$$\left. + \frac{g}{\Omega} \left(\hat{b}_i^\dagger (\hat{n}_j) + \hat{b}_j (\hat{n}_i) \right) + h.c. \right), \tag{A.5b}$$

where $\hat{n}_{i/j}$ is the photon number operator for i^{th}/j^{th} site and $X_j = e^{\frac{g}{\Omega}(\hat{b}_j^\dagger - \hat{b}_j)}$. Terms such as $\frac{g^2}{\Omega^2}(\hat{n}_i \hat{n}_j)$ is taken as zero because we consider only one photon case, and it can not be present at two sites i and j simultaneously. $\hat{b}_i^\dagger(\hat{n}_j)$ is also taken as zero because phonon creation at the ith site for photon present at the jth site is impossible [4].

Now, as before, $\hat{H}'_{om}$ is diagonal, and we can find its eigenvalues and treat $\hat{H}'_{hop}$ as small perturbation. Energy eigenvalues for the jth cavity for m_j number of phonon states and ground-state energy eigenvalue are given as:

$$E_j = (\omega - \frac{g^2}{\Omega}) + \Omega m_j \qquad\qquad E_{GS} = \omega - \frac{g^2}{\Omega} \tag{A.6}$$

Note: We obtain the ground-state energy E_{GS} by diagonolizing the Hamiltonian using the polaron transformation. This is the same as the result we got when we used the polaron ground-state as a coherent state on single-site optomechanical Hamiltonian and calculated the energy eigenvalues through a variational method , see Eq. 3.3. The Unitary transformation does not change eigenvalues.

Eigenstates for $\hat{H}'_{OM}$ for single photon is given by tensor product state $|\, j, m_j \,\rangle$, where $|\, j \,\rangle$ represents photon present at j^{th} site and $|\, m_j \,\rangle$ gives the number of phonon modes at that site. E_{GS} tells about the displacement of mirror due to presence of single photon. Last term in $\hat{H}'_{OM}$ signifies non-linear effects that can be seen for $g > \Omega$ [22].

We can calculate the ground state of $\hat{H}'_{OM}$ by inverse transformation as done before and get the ground state in the original basis as *coherent state* only, see section 2.2.3. $\hat{H}'_{OM}$ produces N-degenerate ground states, as each site can have photon localized there, and $\hat{H}'_{hop}$ lifts the degeneracy.

Treating $\hat{H}'_{hop}$ as a small perturbation and calculating the first order correction for photon and phonon tunneling period using the ground state of phonon and photon excitation at that site, we get,

$$K_{eff} = \langle\, 0, i\, |\hat{b}_i^\dagger \hat{b}_j|\, j, 0\, \rangle = 0 \ , \tag{A.7a}$$

$$J_{eff} = J\ \langle\, 0, i\, |\hat{a}_i^\dagger \hat{a}_j X_i^\dagger X_j|\, j, 0\, \rangle = J e^{-\frac{g^2}{\Omega^2}} \ . \tag{A.7b}$$

Phonon tunneling amplitude K vanishes, and we get exponentially reduced photon tunneling amplitude. We saw the same results when we solved Schrödinger's equation using the polaron basis on optomechanical arrays Hamiltonian in the section 3.2.2.

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